

Quiz 1 – Grading Scheme

Problem 1

$$\begin{aligned} \begin{pmatrix} 0 & 0 & 2 & 4 \\ 1 & 5 & 2 & 1 \\ 1 & 5 & 3 & 3 \end{pmatrix} &\xrightarrow{R_1 \leftrightarrow R_3} \begin{pmatrix} 1 & 5 & 3 & 3 \\ 1 & 5 & 2 & 1 \\ 0 & 0 & 2 & 4 \end{pmatrix} \xrightarrow{R_2 \rightarrow R_2 - R_1} \begin{pmatrix} 1 & 5 & 3 & 3 \\ 0 & 0 & -1 & -2 \\ 0 & 0 & 2 & 4 \end{pmatrix} \xrightarrow{R_3 \rightarrow R_3 + 2R_2} \begin{pmatrix} 1 & 5 & 3 & 3 \\ 0 & 0 & -1 & -2 \\ 0 & 0 & 0 & 0 \end{pmatrix} \\ &\xrightarrow{R_1 \rightarrow R_1 + 3R_2} \begin{pmatrix} 1 & 5 & 0 & -3 \\ 0 & 0 & -1 & -2 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_2 \rightarrow -R_2} \begin{pmatrix} 1 & 5 & 0 & -3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix} \end{aligned}$$

(1 pt. for each row operation.)

Problem 2

(i) Augmented matrix $\rightarrow$ RREF (1.5 marks)

(ii) Translate RREF of $[A \mid b]$ into equations:

$$x_1 + 5x_2 - 3x_4 = b_1, \quad x_3 + 2x_4 = b_2.$$

Write the solution as $x_1 =$, $x_2 =$, $x_3 =$, $x_4 =$, parametrising the free and pivot variables.
(1.5 marks)

(iii) ONE explicit solution to translate with, and explicitly label it. Find explicit p . (1 mark)

(iv) Represent the null space as a span, then explicitly translate. Define W . (1 mark)

Problem 3

Observing that consistency $\iff$ no pivot in the augmented column 1
(last column)

Performing row operations on $[M \mid b]$ with $b = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$ 3

Obtaining the final relation, $b_3 - b_2 - \frac{b_1}{2} = 0$ 1

Problem 4 – Marking Scheme

Case 1 (systematic way)

i) For writing equations (2.5)

ii) For explicit matrix (2.5)

(−1) mark for a “minor” calculation error.

Case 2 (trial and error)

• Correct matrix with justification (5)

• Only correct matrix (slight error) (2.5)

Else, any progress (1)

Problem 5 – Rubric

Part (c)

Full proof

1 point (transpose)

Parts (a) and (b)

- 1) Columns independent $\iff Ax = 0$ has only the trivial solution $\iff \text{REF}(A)$ has a pivot in each column. **3 points**

Deductions (for each):

1. Failing to mention “iff” $-\frac{1}{2}$ point
2. Failing to justify $-\frac{1}{2}$ point

Part (a)

- 2) Row operations leave the row span invariant **1 point**
3) Correct form in REF satisfies the required property **2 points**

Part (b)

- 4) Full proof **3 points**

Partials, depending on quality:

Any progress in a correct direction **1 point**